A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 1: Information Representation

Subtopic 1.1: Data Representation

Concept 1.1.1: Binary vs Decimal Prefixes

Total Questions: 6

Mark Schemes begin on Page 4

Question 1 | Source: 9618_w24_qp_11 (Q1.a)

1 **(a)** State **one** difference between a tebibyte and a gigabyte.

....

.... [1]

Question 2 | Source: 9618_s24_qp_12 (Q7.a)

7 A computer stores binary data.

(a) Tick (✓) **one** box only to identify the **largest** file size.

☐ 3300 kibibytes

☐ 0.3 megabytes

☐ 3 mebibytes

☐ 3300 kilobytes

[1]

Question 3 | Source: 9618_s24_qp_13 (Q1.a)

1 **(a)** Complete the following description.

A kibibyte has a .... prefix. Three kibibytes is the same as .... bytes.

A megabyte has a .... prefix. Two terabytes is the same as .... gigabytes.

[4]

Question 4 | Source: 9618_w23_qp_12 (Q3.a)

3 **(a)** State **one** difference between a kibibyte and a megabyte.

....

.... [1]

Question 5 | Source: 9618_s23_qp_11 (Q3.d.i)

3 A computer has an Operating System (OS).

(d) The computer stores data in binary form.

(i) State the difference between a kibibyte and a kilobyte.

....

.... [1]

Question 6 | Source: 9618_w21_qp_11 (Q1.a)

1 (a) Draw **one** line from each binary value to its equivalent (same) value on the right.

Binary value	
	1 kibibyte
8 bits	1 gigabyte
8000 bits	1 byte
1000 kilobytes	1 kilobyte
1024 mebibytes	1 gibibyte
8192 bits	1 megabyte
	1 mebibyte

[5]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w24_ms_11 (Q1.a)

1(a)	**1 mark** for: • A tebibyte = 1024 gibibytes / 1 048 576 kibibytes / 2^{40} bytes whereas a gigabyte = 1000 megabytes / 1 000 000 kilobytes / 10^9 bytes • Tebi is binary prefix giga is denary prefix	**1**

Mark Scheme for Question 2 | Source: 9618_s24_ms_12 (Q7.a)

7(a)	**1 mark** for: 3300 kibibytes	**1**

Mark Scheme for Question 3 | Source: 9618_s24_ms_13 (Q1.a)

1(a)	**1 mark** for each correct answer: • binary • 3072 • denary/decimal • 2000 A kibibyte has a **binary** prefix. Three kibibytes is the same as **3072** bytes. A megabyte has a **decimal/denary** prefix. Two terabytes is the same as **2000** gigabytes.	**4**

Mark Scheme for Question 4 | Source: 9618_w23_ms_12 (Q3.a)

3(a)	**1 mark** for: Either • kibibyte = 1,024 bytes // 2^{10} bytes **and** megabyte = 1000 kilobytes //1 000 000 bytes // 10^3 kilobytes // 10^6 bytes Or • kibi is binary prefix **and** mega is denary prefix	**1**

Mark Scheme for Question 5 | Source: 9618_s23_ms_11 (Q3.d.i)

3(d)(i)	**1 mark** from • Kibibyte is 1024 bytes and kilobyte is 1000 bytes • Kibibyte is binary prefix and kilobyte is denary prefix	**1**

Mark Scheme for Question 6 | Source: 9618_w21_ms_11 (Q1.a)

1(a)	**1 mark** for each correct line	**5**